

Single-Cell RNA-Seq Analysis: EDA

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I. INTRODUCTION

This paper deals with the exploratory data Analysis and visualization (EDA) of single-cell RNA-sequences to identify patterns and clusters. The perturbation dataset from Norman et al. [1] is used.

II. DATA

The used perturbation dataset contains 91205 cells, from which 7353 are control cells and 83852 are pertubated cells. All cell instances in the dataset are from the A549 cell, a cell used to test the effects of various drugs and treatments on lung cancer cells [2]. The dataset contains 5045 genes and shows how different conditions (perturbations) affect gene expression in individual cells.

III. METHODS

A. PCA

Principal Component Analysis (PCA) was applied to reduce the dimensionality of the gene expression data while preserving the most important patterns. The data was first standardized and then PCA was performed with 128 components.

B. K-Means Clustering

K-means clustering was employed to identify groups of cells with similar gene expression patterns. The optimal number of clusters was determined using the elbow method, which plots the inertia against different numbers of clusters. The analysis was performed on the PCA-reduced data to improve computational efficiency and reduce noise.

C. Violin Plot

Violin plots were generated to visualize the distribution of gene expression levels across different conditions for selected genes. These plots combine box plot and kernel density estimation, showing both the distribution shape and summary statistics.

D. Heatmap

A heatmap was created to visualize the expression patterns of the top variable genes across different conditions. The data were normalized, and genes were selected based on their variance across all samples. The resulting visualization uses a color gradient from dark green (low expression) to yellow (high expression) to represent expression levels, allowing for

the identification of coordinated gene expression patterns and condition-specific responses.

IV. RESULTS

The PCA revealed that the 128 principal components explain 10% of the total variance in the data, indicating the complexity of the gene expression patterns. The cumulative explained variance across all components show a gradual increase with additional components. The first two principal PC1 and PC2 components explain around 1.3% of the variance.

K-means clustering analysis identified nine clusters. These clusters show a strong separation in the PC1 and PC2 space, indicating distinct gene expression patterns and varying responses to perturbations among the cell. Comparing the mean highest and lowest expression of a gene for each cluster, showed significant higher expression of MALAT1 caused by different conditions. The gene CLC showed the highest variance among all clusters and the condition CEBPA is the most prevalent within that cluster.

The violin plot analysis reveals both stable and variable patterns. CKS2 consistently shows low, stable expression, indicating a role in core cellular functions. In contrast, GYP A also displays low expression but with high variability, suggesting sensitivity to specific conditions. TUBB exhibits moderate, consistent expression, implying involvement in steady cellular mechanisms. The condition globally suppresses the expression of all analyzed genes, highlighting its strong inhibitory effect. The heatmap further confirms these observations. MALAT1 is consistently highly expressed across multiple conditions, indicating a potential regulatory role in perturbation-induced processes. CKS2 and GYP A align with their violin plot profiles, emphasizing stability and variability, respectively. Together, these findings highlight dynamic expression patterns, with genes like MALAT1 and GYP A as potential targets for further study of condition-specific gene regulation.

REFERENCES

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